COMPREHENSIVE LESSON GUIDE & STUDENT WORKSHEET

# Graph Theory: Networks & Shortest Path

Special Edition: The Great Indian Wedding Express & Practical Network Optimization

**Scenario:** A grand wedding is happening in Delhi! The *Wedding Express* train must rush guests across the city to reach various marriage venues and stations on time. But with multiple tracks and delays, which route is guaranteed to be the fastest? We use **Graph Theory** to navigate and optimize real-world networks!

## 1. Core Components: What are Nodes and Edges?

In graph theory, any complex system or network is built using two fundamental elements:

| Component | Graph Theory Definition | Real-World Transportation Example |
|---|---|---|
| **Node (Vertex)** | A distinct point, junction, or location within a system. Represented as dots in a graph. | A railway station (e.g., New Delhi Railway Station), metro station, landmark, or delivery hub. |
| **Edge (Link)** | A connection or line between two distinct nodes that allows movement or flow. | A direct railway track, road segment, flight path, or underground metro line. |
| **Graph (Network)** | A collection of nodes connected by edges: $G = (V, E)$. | The complete Delhi Metro network or the entire Indian Railways transit web. |

## 2. Unweighted vs. Weighted Networks

Networks can either show simple connectivity or provide detailed quantitative measurement metrics:

| Network Type | Key Characteristics | Practical Example |
|---|---|---|
| **Unweighted Network** | Edges only show *whether* two nodes are connected. All paths look equal in length or difficulty; no numerical values are attached. | A basic transit map showing which metro line connects to which station, without travel duration info. |
| **Weighted Network** | Edges carry numerical values called **weights** representing physical attributes such as travel time, distance, or financial cost. | Travel time in minutes between stations (e.g., 4 minutes from Rajiv Chowk to Patel Chowk). |
| **Directed Network (Digraph)** | Edges have arrows indicating one-way flow/ direction between nodes. | One-way streets, flight schedules, or pipeline flows. |

## 3. Mathematical Foundations & Shortest Path Algorithm

To systematically find the shortest, fastest, or cheapest route in a weighted network, we follow a standardized algorithmic approach (akin to Dijkstra's algorithm logic):

**STEP 1** **Identify All Paths:** Enumerate every valid, non-repeating path from the starting source node ($S$) to the target destination node ($T$).

**STEP 2** **Calculate Total Weight:** Sum up the edge weights (time, distance, or cost) along each identified route.

**STEP 3** **Compare & Select:** Find the minimum total value. The route corresponding to this minimum sum is the optimal *Shortest Path*.

**Mathematical Expression:** For a path $P = (v_1, v_2, ..., v_k)$, the path weight $W(P) = \sum w(v_i, v_{i+1})$. We seek $P^*$ such that $W(P^*) = \min [W(P)]$.

## 4. Worked Examples & Case Studies

### Case Study 1: Mumbai Dabbawala Lunch Delivery Network

The Dabbawalas deliver thousands of lunch boxes daily across Mumbai with precision. Consider a local delivery sub-network connecting 5 stops: $A, B, C, D, E$.

**Given Edge Weights (Travel Time in Minutes):**

- $A \to B = 4\ min$, $A \to C = 2\ min$
- $B \to D = 5\ min$, $B \to E = 8\ min$
- $C \to D = 3\ min$, $C \to E = 9\ min$
- $D \to E = 2\ min$

**Route Evaluation (Finding Fastest Route from A to E):**

| Route Option | Path Breakdown | Calculation | Total Time |
|---|---|---|---|
| Route 1 | $A \to B \to E$ | $4 + 8$ | 12 minutes |
| Route 2 | $A \to B \to D \to E$ | $4 + 5 + 2$ | 11 minutes |
| Route 3 | $A \to C \to E$ | $2 + 9$ | 11 minutes |
| **Route 4** | $A \to C \to D \to E$ | $2 + 3 + 2$ | **7 minutes** |

**Optimal Path: A → C → D → E (Total Time: 7 minutes)**

### Case Study 2: School Bus Route Optimizer

Find the fastest school bus route from Home ($A$) to School ($D$) passing through Park ($B$) or Market ($C$).

**Given Weights:** $AB = 4$, $AC = 7$, $BD = 5$, $CD = 3$, $BC = 2$ (all in min).

**Route Evaluation:**

- **Route 1** ($A \to B \to D$): $4 + 5 = 9 ext{ min}$
- **Route 2** ($A \to C \to D$): $7 + 3 = 10 ext{ min}$
- **Route 3** ($A \to B \to C \to D$): $4 + 2 + 3 = 9 ext{ min}$

**Result: TIE between Route 1 (A → B → D) and Route 3 (A → B → C → D) at 9 minutes!**

### Case Study 3: The Wedding Express - Delhi Station Rush

Guests need to travel from Chandni Chowk ($S$) to the Wedding Venue near Airport Terminal ($T$). Intermediary hubs are Rajiv Chowk ($R$), Hauz Khas ($H$), and Aerocity ($A$).

**Given Times:** $S \to R = 8\ min$, $S \to H = 18\ min$, $R \to H = 10\ min$, $R \to A = 22\ min$, $H \to A = 8\ min$, $A \to T = 5\ min$, $H \to T = 16\ min$.

**Route Evaluation:**

- **Route 1** ($S \to R \to A \to T$): $8 + 22 + 5 = 35$ ext{ min}
- **Route 2** ($S \to R \to H \to A \to T$): $8 + 10 + 8 + 5 = 31$ ext{ min}
- **Route 3** ($S \to H \to A \to T$): $18 + 8 + 5 = 31$ ext{ min}
- **Route 4** ($S \to H \to T$): $18 + 16 = 34$ ext{ min}

**Optimal Path: Routes 2 and 3 both achieve the fastest time of 31 minutes.**

## 5. Real-World Applications Across Infrastructure

| Urban Transit & Metro Networks | Logistics & Express Delivery | National Highway Planning |
|---|---|---|
| Routing algorithms like Dijkstra's & A* powers navigation apps (Google Maps, Delhi Metro App) to compute instant transfer routes. | E-commerce (Amazon, Delhivery) and Mumbai Dabbawalas optimize last-mile fuel usage and transit timelines. | Golden Quadrilateral & Expressways optimize freight corridors between major industrial hubs. |

## 6. Extended Practice Worksheet & Student Exercises

Complete the following multi-part practice problems to demonstrate your understanding of network modeling and shortest path calculations.

### Question 1: Foundational Terminology & Concepts

In a city bus routing system, what do the bus stops represent and what do the roads connecting them represent in Graph Theory terms? Explain the critical difference between an unweighted network and a weighted network in this specific context.

*Student Workspace / Notes:*

### Question 2: Delivery Network Shortest Path Calculation

A delivery driver needs to go from Warehouse (Node $S$) to Customer (Node $T$). The network graph edges and travel times in minutes are as follows:

- $S \rightarrow A = 3\ ext\{\ min\} \mid S \rightarrow B = 6\ ext\{\ min\}$
- $A \rightarrow B = 1\ ext\{\ min\} \mid A \rightarrow T = 8\ ext\{\ min\}$
- $B \rightarrow T = 3\ ext\{\ min\}$

**Task:** List all possible paths from $S$ to $T$, calculate the total travel time for each path, and identify the single shortest path.

*Path Breakdown & Calculations:*

### Question 3: Metro Route Trade-off Comparison

Suppose you are comparing two metro routes between two distant stations:

- **Route X:** Has 3 intermediate stops with segment travel times of 5, 4, and 6 minutes.
- **Route Y:** Has only 1 intermediate stop, but due to severe congestion, the two segments take 9 and 7 minutes respectively.

Which route is the shortest path in terms of total time? Show your complete mathematical work and explain why having fewer stops does not always mean a faster journey.

*Workspace & Explanation:*

### Question 4: E-Commerce Freight Hub Optimization (Advanced)

A logistics company transfers cargo from Hub $P$ to Hub $Z$ through intermediate fulfillment centers $Q, R, S, T$. The travel times (in hours) are given as:

$$P \rightarrow Q = 4 \mid P \rightarrow R = 7 \mid Q \rightarrow S = 5 \mid Q \rightarrow R = 2 \mid R \rightarrow T = 3 \mid S \rightarrow Z = 4 \mid T \rightarrow Z = 2 \mid S \rightarrow T = 1$$

**Task:** Determine the optimal path with the minimum travel time from $P$ to $Z$. Show step-by-step evaluations for at least 4 distinct routes.

*Detailed Step-by-Step Path Evaluation:*

### Question 5: Real-World Neighborhood Network Design

Select 4 actual or fictional places in your neighborhood (e.g., Home, School, Library, Bakery). Draw/define the network below, assign realistic time weights (in minutes) to each connecting road/edge, and show how you would calculate the fastest path between your Home and the Library.

*Network Map Representation & Path Computation:*

### Summary Key Takeaways:

- Nodes represent locations/points; Edges represent links/pathways connecting them.
- Weighted edges assign quantitative metrics (time, distance, cost) critical for real-world decision making.
- The Shortest Path is the sequence of edges connecting two nodes that minimizes total cumulative weight.
- Graph theory powers modern navigation, telecommunications, logistics, and supply chain infrastructure across India and globally.